DDA3020 Homework 2 - Solution

April 7, 2025

1 Written Problems (50 points)

1.1. (Decision Trees, 30 points)

(2 points)Question 1: In a decision tree, what is the purpose of the leaf nodes?

A) To represent the class label or value to be predicted

B) To store the conditions for splitting the data

C) To indicate the importance of a feature

D) To represent the depth of the tree

Answer: A

(2 points)Question 2: Which of the following algorithms can be used for both classification and regression tasks?

A) Decision trees

B) Logistic Regression

C) Support Vector Machines

D) All of the above

Answer: A or AC

A must be right, whether C chooses or not is considered correct. Strictly speaking, SVR is the support vector machine used for regression. .

(2 points)Question 3: How can decision trees be made more robust to noise in the data?

A) By increasing the maximum depth of the tree

B) By using a smaller minimum samples per leaf

C) By using ensemble techniques like bagging or boosting

D) By removing features with low importance

Answer: C

(2 points)Question 4: How do decision trees handle continuous variables?

A) By discretizing the continuous variables into intervals

B) By using one-hot encoding

C) By normalizing the continuous variables

D) By ignoring the continuous variables

Answer: A

(2 points)Question 5: What is entropy in the context of decision trees?

A) A measure of disorder or impurity in a node

B) A measure of the complexity of a decision tree

C) The difference between the predicted and actual values in a node

D) The rate at which information is gained in a decision tree

Answer: A

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Take a look at the following data. Here, the Y label indicates whether a child goes out to play or not.

1. (12 points, 3 points for each attribute, two decimal places are sufficient) Calculate the information gain for each attribute and select the optimal splitting attribute (to construct the first layer of the tree).

   Answer:

$$
\begin{aligned}
H(Y \mid \text{weather}) &= -\sum_x P(\text{weather} = x) \sum_y P(Y = y \mid \text{weather} = x) \log_2 P(Y = y \mid \text{weather} = x) \\
&= -\Bigg(P(\text{weather} = S) \sum_y P(Y = y \mid \text{weather} = S) \log_2 P(Y = y \mid \text{weather} = S) + \\
&\quad P(\text{weather} = C) \sum_y P(Y = y \mid \text{weather} = C) \log_2 P(Y = y \mid \text{weather} = C) + \\
&\quad P(\text{weather} = R) \sum_y P(Y = y \mid \text{weather} = R) \log_2 P(Y = y \mid \text{weather} = R) \Bigg) \\
&= -\left((0.3) \left(\left(\frac{1}{3}\right) \log_2 \left(\frac{1}{3}\right) + \left(\frac{2}{3}\right) \log_2 \left(\frac{2}{3}\right) \right) + \right. \\
&\quad (0.3) \left((1) \log_2 (1) + (0) \log_2 (0) \right) + \\
&\quad \left. (0.4) \left(\left(\frac{1}{4}\right) \log_2 \left(\frac{1}{4}\right) + \left(\frac{3}{4}\right) \log_2 \left(\frac{3}{4}\right) \right) \right) \\
&= 0.6 \\
\text{IG}(Y, \text{weather}) &= 1 - 0.6 = 0.4
\end{aligned}
$$

$$
\begin{aligned}
H(Y \mid temp) &= -\sum_x P(temp = x) \sum_y P(Y = y \mid temp = x) \log_2 P(Y = y \mid temp = x) \\
&= -\Bigg(P(temp = H) \sum_y P(Y = y \mid temp = H) \log_2 P(Y = y \mid temp = H) + \\
&\quad P(temp = M) \sum_y P(Y = y \mid temp = M) \log_2 P(Y = y \mid temp = M) + \\
&\quad P(temp = C) \sum_y P(Y = y \mid temp = C) \log_2 P(Y = y \mid temp = C) \Bigg) \\
&= -\left((0.4) \left(\left(\frac{1}{2}\right) \log_2 \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) \log_2 \left(\frac{1}{2}\right) \right) + \right. \\
&\quad (0.5) \left(\left(\frac{3}{5}\right) \log_2 \left(\frac{3}{5}\right) + \left(\frac{2}{5}\right) \log_2 \left(\frac{2}{5}\right) \right) + \\
&\quad \left. (0.1) \left((1) \log_2 (1) + (0) \log_2 (0) \right) \right) \\
&= 0.8855 \\
\text{IG}(Y, \text{temp}) &= 1 - 0.8855 = 0.1145
\end{aligned}
$$

$$
\begin{aligned}
H(Y \mid \text{hum}) &= -\sum_x P(\text{hum} = x) \sum_y P(Y = y \mid \text{hum} = x) \log_2 P(Y = y \mid \text{hum} = x) \\
&= -\Bigg(P(\text{hum} = H) \sum_y P(Y = y \mid \text{hum} = H) \log_2 P(Y = y \mid \text{hum} = H) + \\
&\quad P(\text{hum} = N) \sum_y P(Y = y \mid \text{hum} = N) \log_2 P(Y = y \mid \text{hum} = N) \Bigg) \\
&= -\left((0.7) \left(\left(\frac{3}{7}\right) \log_2 \left(\frac{3}{7}\right) + \left(\frac{4}{7}\right) \log_2 \left(\frac{4}{7}\right) \right) + \right. \\
&\quad \left. (0.3) \left(\left(\frac{2}{3}\right) \log_2 \left(\frac{2}{3}\right) + \left(\frac{1}{3}\right) \log_2 \left(\frac{1}{3}\right) \right) \right) \\
&= 0.9642 \\
\text{IG}(Y, \text{hum}) &= 1 - 0.9642 = 0.0358
\end{aligned}
$$

$$
\begin{aligned}
H(Y \mid \text{wind}) &= -\sum_x P(\text{wind}=x)\sum_y P(Y=y \mid \text{wind}=x)\log_2 P(Y=y \mid \text{wind}=x) \\
&= -\Bigg(P(\text{wind}=S)\sum_y P(Y=y \mid \text{wind}=S)\log_2 P(Y=y \mid \text{wind}=S)+ \\
&\quad P(\text{wind}=W)\sum_y P(Y=y \mid \text{wind}=W)\log_2 P(Y=y \mid \text{wind}=W)\Bigg) \\
&= -\left((0.6)\left(\left(\frac{2}{6}\right)\log_2\left(\frac{2}{6}\right)+\left(\frac{4}{6}\right)\log_2\left(\frac{4}{6}\right)\right)+\right. \\
&\quad \left.(0.4)\left(\left(\frac{1}{4}\right)\log_2\left(\frac{1}{4}\right)+\left(\frac{3}{4}\right)\log_2\left(\frac{3}{4}\right)\right)\right) \\
&= 0.8755 \\
\mathrm{IG}(Y, \text{wind}) &= 1 - 0.8755 = 0.1245
\end{aligned}
$$

Since "weather" has the biggest information gain, it will be the optimal splitting attribute for the first layer of the tree.

2. (8 points) Draw the final tree structure without writing out the calculation process. (Hint: The error rate of the final training set is 0, that is, all samples are correctly classified.)

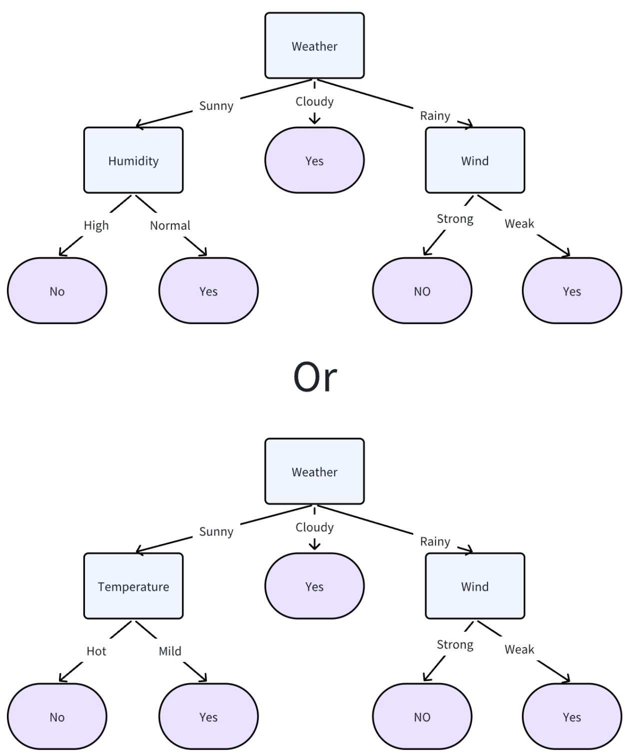

Figure 1: Answer of draw the tree

1.2.(CNN, 20 points) Consider the convolutional network defined by the layers below. The input shape is $28 \times 28 \times 1$ and the output is 5 neurons. Consider layers:

$$\text{Conv4}(15) + \text{Maxpool}_2 + \text{Conv2}(25) + \text{Maxpool}_2 + \text{FC5}$$

where

- Conv4(15): 15 filters with each size $4 \times 4 \times D$, where D is the depth of the activation volume at the previous layer, stride = 1, padding = 1;
- Conv2(25): 25 filters with each size $2 \times 2 \times D$, where D is the depth of the activation volume at the previous layer, stride = 1, padding = 0;
- Maxpool_2: 2×2 filter, stride = 2, padding = 0;
- FC5: A fully-connected layer with 5 output neurons.

1. (10 pts) Compute the shape of activation map of each layer.
2. (10 pts) Compute the total number of parameters of each layer.

The answer is as in Table 2.

Table 1: Understanding and calculating the number of parameters in CNN

NO.	**Layer**	**Activation Shape**	**# Parameters**	Mark
1	Input Layer	(28,28,1)	0	-
2	Conv4(15)	(27,27,15)	(4*4*1 + 1)*15 = 255	4 pts
3	Maxpool2	(13,13,15)	0	4 pts
4	Conv2(25)	(12,12,25)	(2*2*15 + 1)*25 = 1525	4 pts
5	Maxpool2	(6,6,25)	0	4 pts
6	FC5	(5,1)	6*6*25*5 + 5 = 4505	4 pts

The following answer is also correct (padding when it cannot be divided).

Table 2: Understanding and calculating the number of parameters in CNN

NO.	**Layer**	**Activation Shape**	**# Parameters**	Mark
1	Input Layer	(28,28,1)	0	-
2	Conv4(15)	(27,27,15)	(4*4*1 + 1)*15 = 255	4 pts
3	Maxpool2	(14,14,15)	0	4 pts
4	Conv2(25)	(13,13,25)	(2*2*15 + 1)*25 = 1525	4 pts
5	Maxpool2	(7,7,25)	0	4 pts
6	FC5	(5,1)	7*7*25*5 + 5 = 6130	4 pts